

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

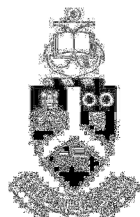
Kopiereg voorbehou

Examination

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Module EBN111
Elektrisiteit en Elektronika
17 Junie 2014

Module EBN111
Electricity and Electronics
17 June 2014



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:*

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		33 (Antwoordblad ingesluit / <i>Answer sheet included</i>)	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Die Engelse teks van hierdie vraestel sal voorkeur geniet met betrekking tot die interpretasie van die inhoud.
The English text of this question paper will be given precedence in the interpretation of the content.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$**
Polar form: $A\angle\theta$; Rectangular form: $a + jb$
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Interne Eksaminator(e): <i>Internal examiner(s):</i>	D. Bhatt
	J.D. le Roux
Eksterne eksaminator: <i>External examiner:</i>	J. Joubert

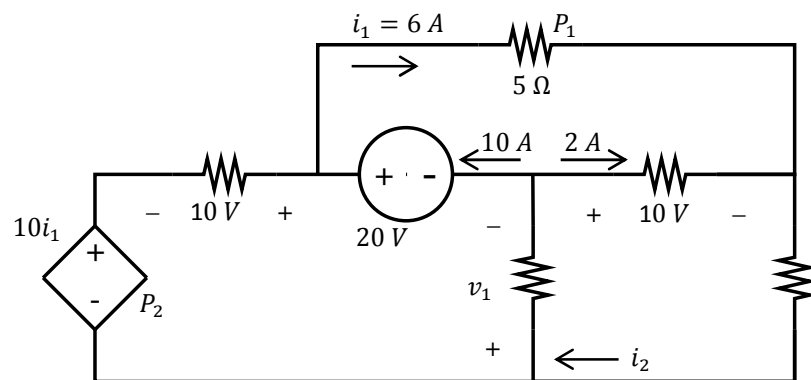
VRAAG/QUESTION 1 [20]

Vraag 1.1 [6]

- a) 'n Huis het twintig ou 100 W tungsten-gloeilampe wat almal vir 12 uur per dag brand. Hoeveel kan die eienaars van die huis spaar per jaar (365 dae) indien die gloeilampe vervang word met nuwe 20 W gloeilampe? Elektrisiteit kos R5/kWh. [2]
- b) 'n 9 V battery word gebruik om 'n radio beheerde kar aan te dryf. Die drywing wat die battery kan lewer val lineêr van 5 W na 2 W in 100 minute. Bepaal die totale energie wat deur die kar verbruik word. [2]
- c) 'n Selfoonbattery lewer 3.024 kJ vir 24 ure aan 'n selfoon teen 'n stroom van 10 mA. Bepaal die spanning van die battery. [2]

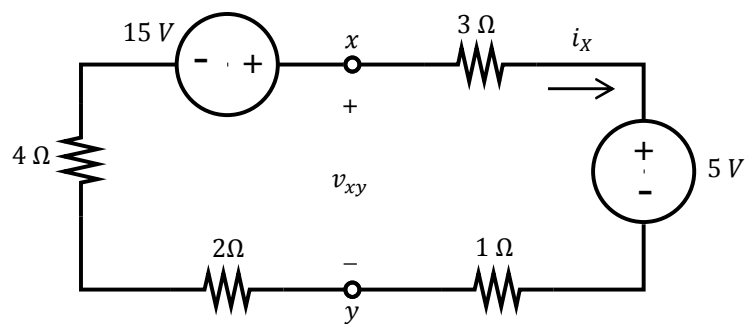
Question 1.1 [6]

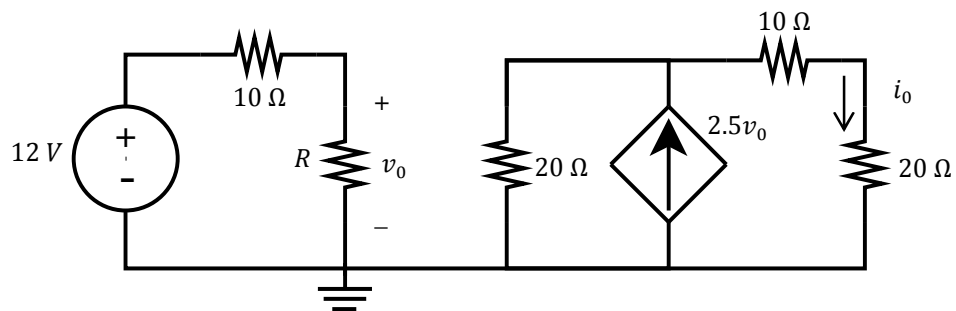
- a) A house has twenty old 100 W tungsten lightbulbs which are switched on for 12 hours per day. How much money can the owners of the house save per year (365 days) if they replace the globes with new 20 W lightbulbs? Electricity costs R5/kWh. [2]
- b) A 9 V battery is used to power a remote controlled car. The power delivered by the battery falls linearly from 5 W to 2 W in 100 minutes. Determine the total energy used by the car. [2]
- c) A cellphone battery provides 3.024 kJ for 24 hours to a cellphone at a current of 10 mA. Determine the voltage of the battery. [2]

Vraag 1.2 [4]Bepaal P_1 , P_2 , v_1 en i_2 .**Question 1.2 [4]**Determine P_1 , P_2 , v_1 and i_2 .

Vraag 1.3 [2]
Bepaal v_{xy} en i_X .

Question 1.3 [2]
Determine v_{xy} and i_X .



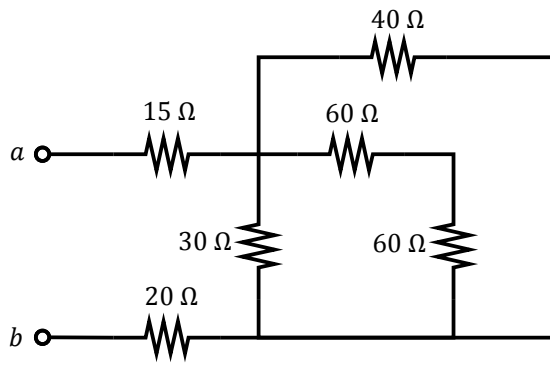
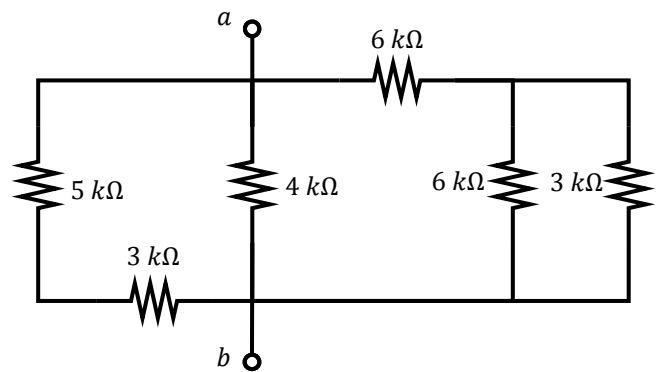
Vraag 1.4 [2]Bepaal R en i_0 , indien $v_0 = 4$ V.**Question 1.4 [2]**Determine R and i_0 , if $v_0 = 4$ V.

Vraag 1.5 [4]

Bepaal die ekwivalente weerstand van onderstaande twee kringbane, ten opsigte van terminale a en b .

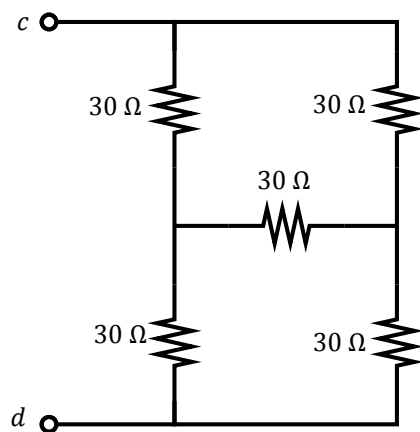
Question 1.5 [4]

Determine the equivalent resistance of the two circuits below with respect to terminals a and b .

Kringbaan / Circuit 1**Kringbaan / Circuit 2**

Vraag 1.6 [2]
Bepaal R_{cd} .

Question 1.6 [2]
Determine R_{cd} .



VRAAG/QUESTION 2 [32]

Vraag 2.1 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$av_1 + bv_2 = 13 \text{ A}$$

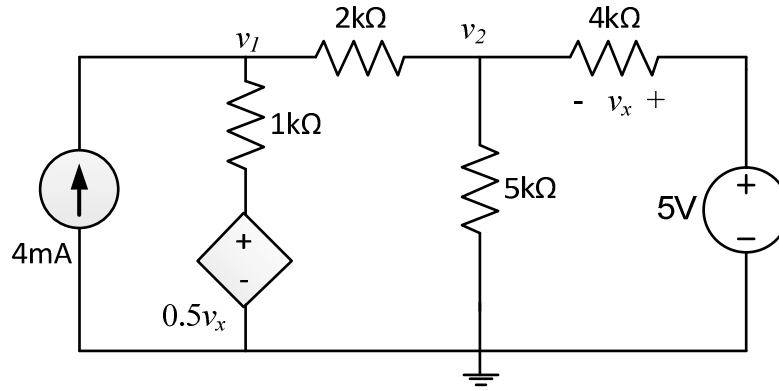
$$cv_1 - 19v_2 = d \text{ A}$$

Question 2.1 [4]

Use nodal analysis to set up two equations of the following form:

$$av_1 + bv_2 = 13 \text{ A}$$

$$cv_1 - 19v_2 = d \text{ A}$$



Vraag 2.2 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$100i_1 + ei_2 = f \text{ A}$$

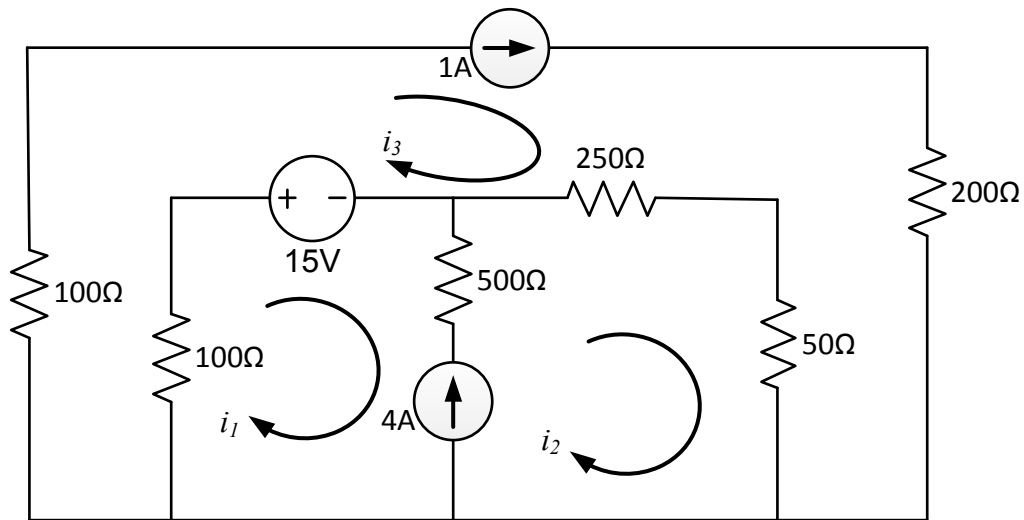
$$gi_1 + hi_2 = -4 \text{ A}$$

Question 2.2 [4]

Use mesh analysis to set up two equations of the following form:

$$100i_1 + ei_2 = f \text{ A}$$

$$gi_1 + hi_2 = -4 \text{ A}$$



Vraag 2.3 [2]

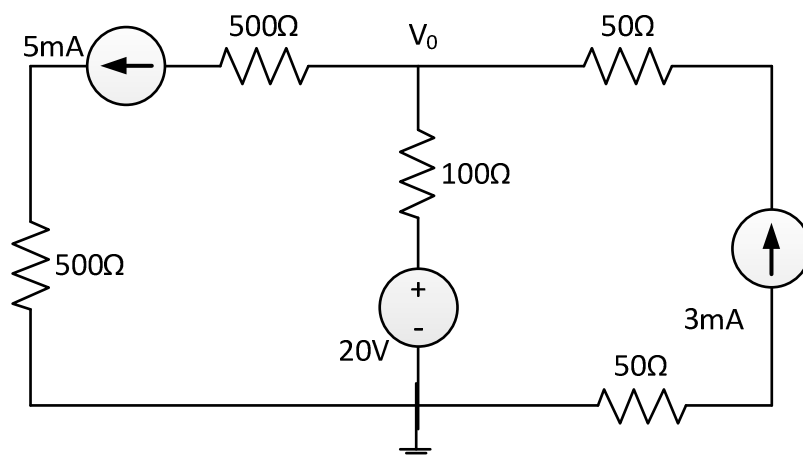
Bepaal m en V indien $I = 1$ A.

$$\begin{bmatrix} 5 & 8 \\ -1 & m \end{bmatrix} \begin{bmatrix} I \\ V \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Question 2.3 [2]

Determine m and V if $I = 1$ A.

$$\begin{bmatrix} 5 & 8 \\ -1 & m \end{bmatrix} \begin{bmatrix} I \\ V \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

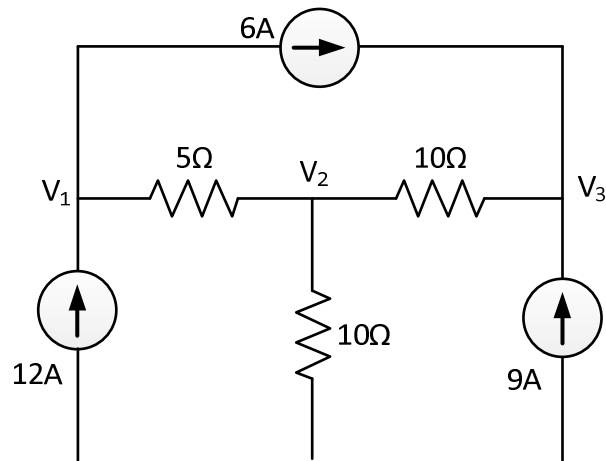
Vraag 2.4 [2]Bepaal die node spanning V_0 .**Question 2.4[2]**Determine the nodal voltage V_0 .

Vraag 2.5 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die lusstrome.

Question 2.5 [3]

Use inspection to set up a matrix equation in terms of the node voltages.



Vraag 2.6 [4]

Gegee: $\beta = 99$; $V_{BE} = 0.7 \text{ V}$. Bepaal:

a) I_E wanneer $v_s = 5 \text{ V}$. (Gee antwoord in mA)

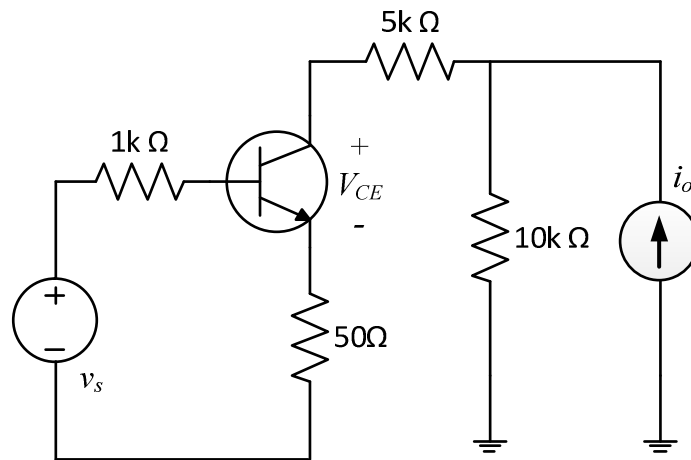
b) I_B wanneer $i_o = 2 \text{ mA}$ en $V_{CE} = 15 \text{ V}$. (Gee antwoord in μA)

Question 2.6 [4]

Given: $\beta = 99$; $V_{BE} = 0.7 \text{ V}$. Calculate:

a) I_E when $v_s = 5 \text{ V}$ (Give answer in mA).

b) I_B when $i_o = 2 \text{ mA}$ and $V_{CE} = 15 \text{ V}$. (Give answer in μA).

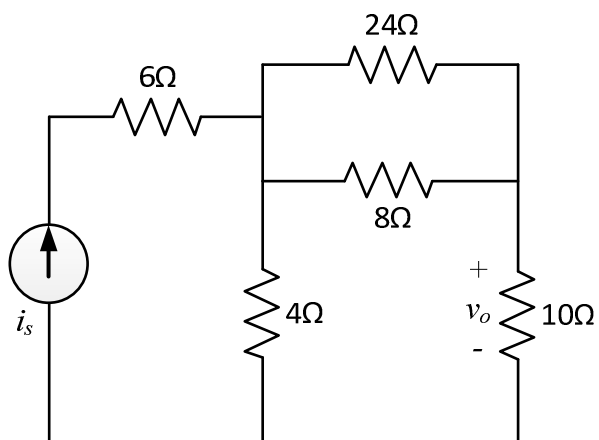


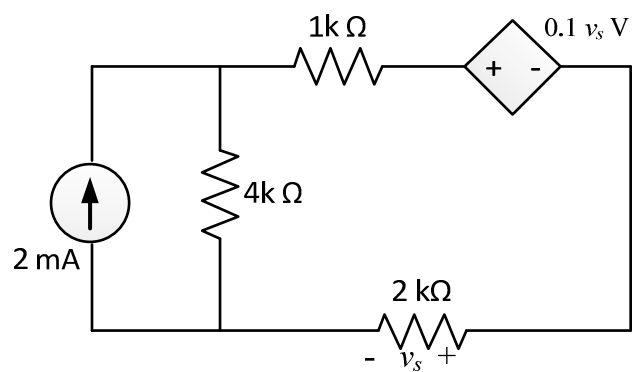
Vraag 2.7 [2]Gebruik die lineariteit beginsel en bepaal v_o wanneer:

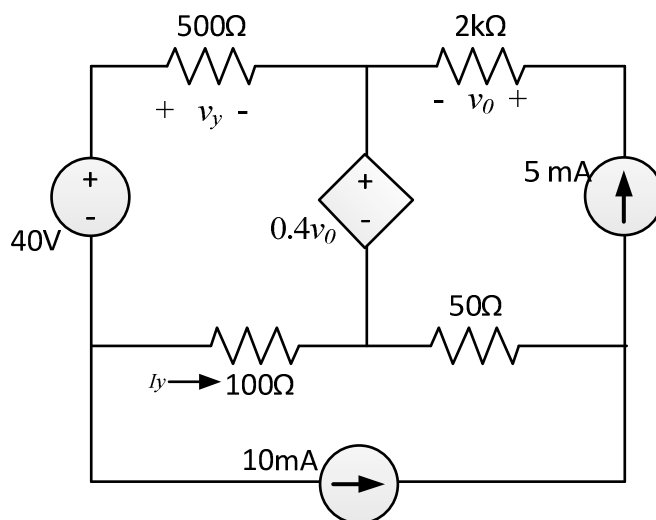
- a) $i_s = 2 \text{ A}$
- b) $i_s = 8 \text{ mA}$

Question 2.7 [2]Using the linearity principle, calculate v_o when:

- a) $i_s = 2 \text{ A}$
- b) $i_s = 8 \text{ mA}$



Vraag 2.8 [2]Maak gebruik van brontransformatie en bepaal v_s .**Question 2.8 [2]**Use source transformation to determine v_s .

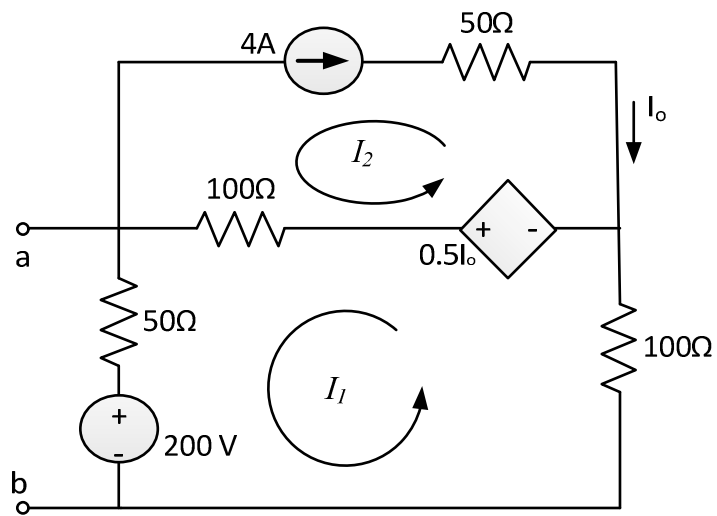
Vraag 2.9 [2]Bepaal die bydrae van die 5m A spanningsbron tot v_y .**Question 2.9 [2]**Calculate the contribution of 5m A source to v_y .

Vraag 2.10 [4]

Bepaal die Thevenin ekwivalent V_{TH} en R_{TH} met betrekking tot terminale a - b .

Question 2.10 [4]

Determine the Thevenin equivalent V_{TH} and R_{TH} with respect to terminals a - b .

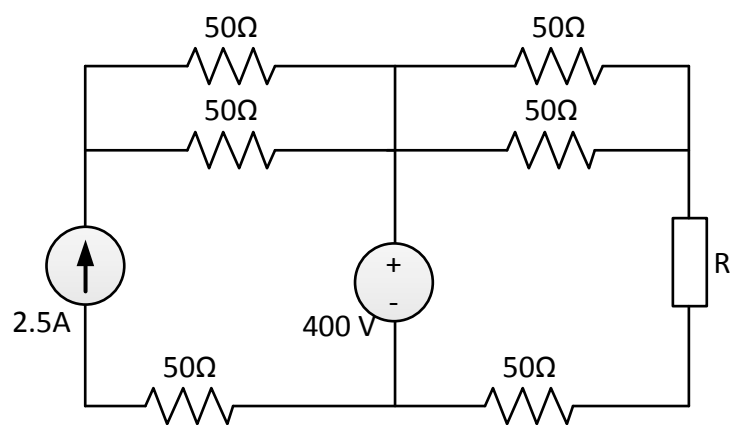


Vraag 2.11 [4]

Bepaal die Norton ekwivalent I_N en R_N met betrekking tot terminale las R .

Question 2.11 [4]

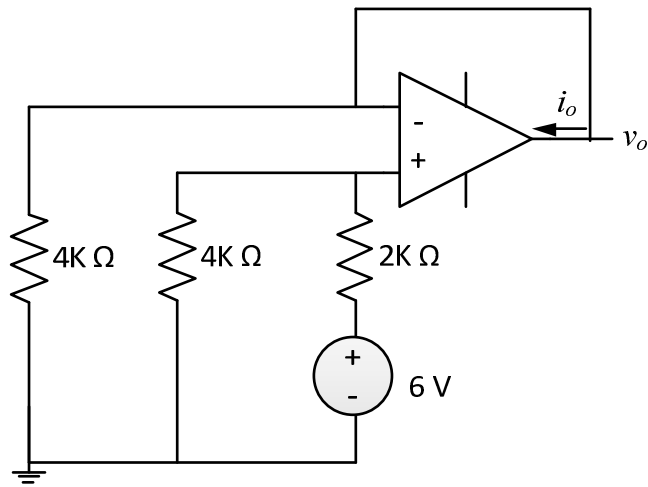
Determine the Norton equivalent current I_N and resistance R_N with respect to load R .



VRAAG/QUESTION 3 [17]

Vraag 3.1 [2]
Bepaal v_o en i_o .

Question 3.1 [2]
Determine v_o and i_o .

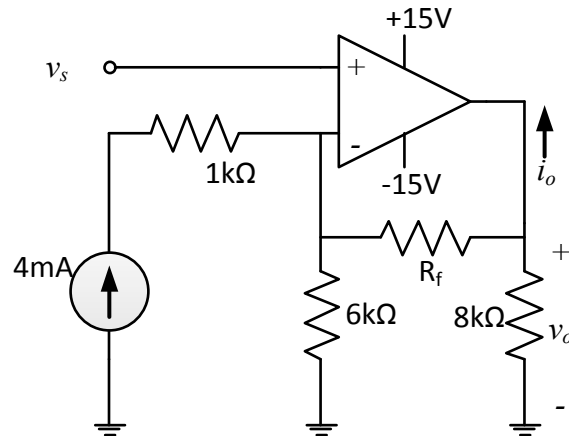


Vraag 3.2 [3]

- a) Watter waarde vir die terugvoer weerstand, R_f , sal veroorsaak dat die op-amp negatief versadig ($-15 \leq v_o \leq 15$) indien $v_s = 3V$?
- b) Wanneer $v_s = 12V$ en $R_f = 2\text{ k}\Omega$, bepaal die stroom i_o en die spanning v_o .

Question 3.2 [3]

- a) What value of feedback resistance, R_f , will negatively saturate ($-15 \leq v_o \leq 15$) the op amp, given $v_s = 3V$?
- b) When $v_s = 12V$ and $R_f = 2\text{ k}\Omega$ calculate the current i_o and voltage v_o .

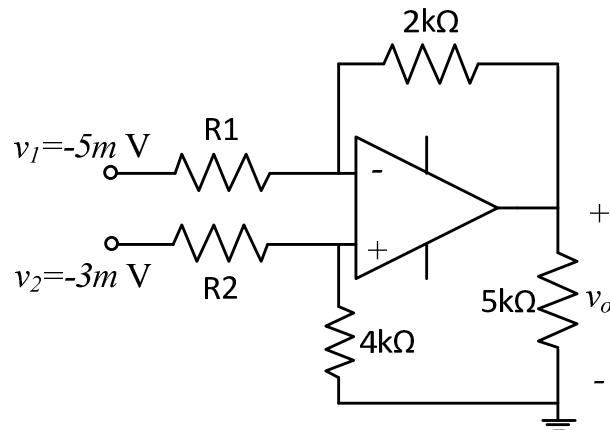


Vraag 3.3 [3]

- a) Gegee: $R_1 = 1\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$. Bepaal v_o .
b) Watter waarde van R_1 en R_2 sal hierdie verskil versterker verander na 'n kringbaan waar $v_o = v_2 - v_1$?

Question 3.3 [3]

- a) Given $R_1 = 1\text{ k}\Omega$, $R_2 = 2\text{ k}\Omega$. Find v_o .
b) What value of R_1 and R_2 will change this difference amplifier into a subtractor ($v_o = v_2 - v_1$)?

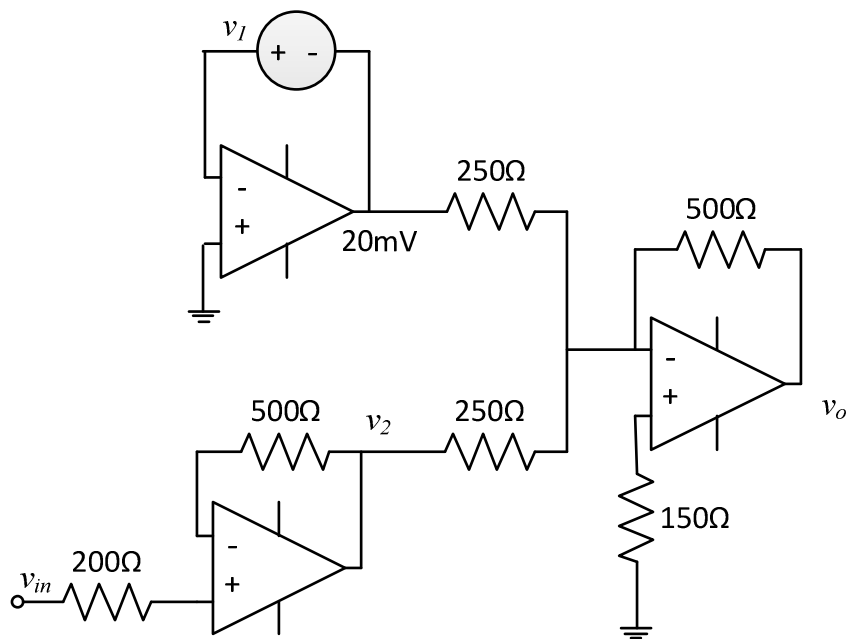


Vraag 3.4 [3]

Dit word gegee dat $v_{in} = -2\text{mV}$. Bepaal v_I , v_2 en v_o

Question 3.4 [3]

It is given that $v_{in} = -2\text{mV}$. Calculate v_I , v_2 and v_o

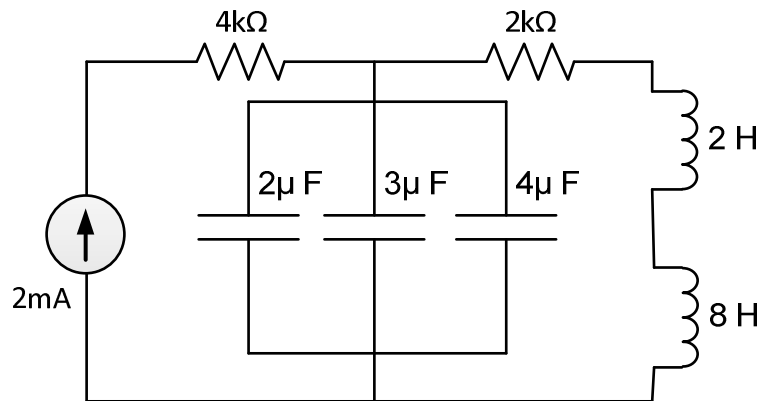


Vraag 3.5 [2]

Aanvaar gelykstroom toestande in die kringbaan. Bepaal die totale energie gestoor in die induktors (w_L) en in die kapasitors (w_C).

Question 3.5 [2]

Assume direct current conditions in the circuit. Determine the total energy stored in the inductors (w_L) and the capacitors (w_C).



Vraag 3.6 [4]

By $t=0.1s$ bepaal die spanning oor die:

a) kapasitors, v_C in figuur (a). Aanvaar $v_C(0) = 0$. Gee die antwoord in mV .

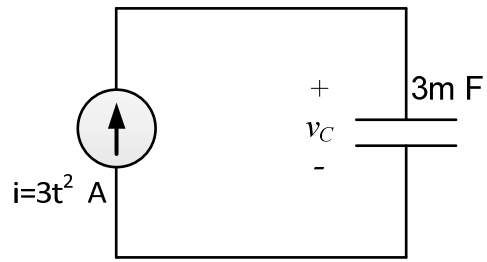
b) induktors v_L in figuur (b). Gee die antwoord in μV .

Question 3.6 [4]

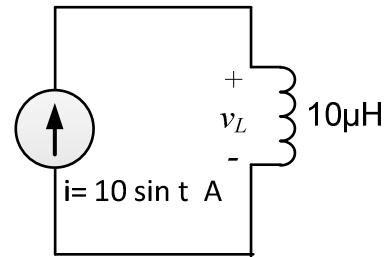
At $t=0.1s$, determine the voltage across the:

a) capacitors, v_C in figure (a). Assume $v_C(0) = 0$. Give the answer in mV .

b) inductors v_L in figure (b). Give the answer in μV .



(a)



(b)

VRAAG/QUESTION 4 [20]

Vraag 4.1 [4]

$$v_1 = -10 \cos(10t - 30^\circ) \text{ V}$$
$$v_2 = 10 \sin(10t + 120^\circ) \text{ V}$$

- a) Bepaal $v_0 = v_1 + v_2$ en skryf die finale antwoord in die vorm $A \cos(\omega t + \phi)$ waar $A > 0$ en $-180^\circ < \phi < 180^\circ$.
- b) Watter een van die sinusvormige golwe v_1 en v_2 loop voor en met hoeveel grade?

Question 4.1 [4]

$$v_1 = -10 \cos(10t - 30^\circ) \text{ V}$$
$$v_2 = 10 \sin(10t + 120^\circ) \text{ V}$$

- a) Determine $v_0 = v_1 + v_2$ and write the final answer in the form $A \cos(\omega t + \phi)$ where $A > 0$ and $-180^\circ < \phi < 180^\circ$.
- b) Which one of the sinusoids v_1 and v_2 are leading and with how many degrees?

Vraag 4.2 [4]a) Vereenvoudig en skryf die antwoord in **polêre vorm**:

$$\bar{I}_1 = \frac{\sqrt{2}(-6 + j4)(5 - j)}{-1 + j\sqrt{3}}$$

b) Vereenvoudig en skryf die antwoord in **polêre vorm**:

$$\bar{V}_1 = \frac{\sqrt{50} \angle 60^\circ}{(-5 + j5)^*}$$

Question 4.2 [4]a) Simplify and write the answer in **polar form**:

$$\bar{I}_1 = \frac{\sqrt{2}(-6 + j4)(5 - j)}{-1 + j\sqrt{3}}$$

b) Determine $\bar{V}_1$ in **polar form**:

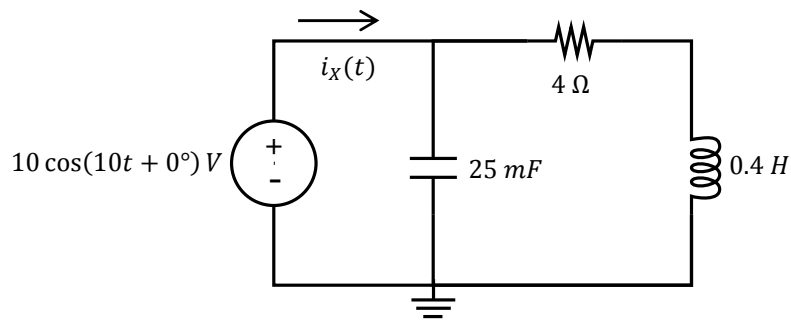
$$\bar{V}_1 = \frac{\sqrt{50} \angle 60^\circ}{(-5 + j5)^*}$$

Vraag 4.3 [2]

Bereken $i_X(t)$. Skryf die finale antwoord in die vorm:
 $i_X(t) = A \cos(\omega t + \phi)$ A met $A > 0$ en $-180^\circ < \phi < 180^\circ$.

Question 4.3 [2]

Determine $i_X(t)$. Write the final answer in the form: $i_X(t) = A \cos(\omega t + \phi)$ A with $A > 0$ and $-180^\circ < \phi < 180^\circ$.

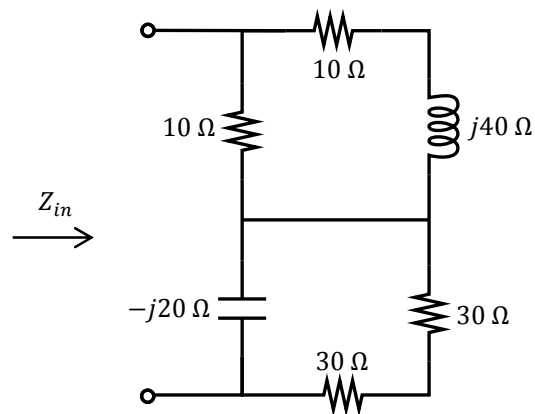


Vraag 4.4 [2]

Bereken Z_{in} en skryf die antwoord in **reghoekige** formaat.

Question 4.4 [2]

Determine Z_{in} and write the answer in **rectangular** format.

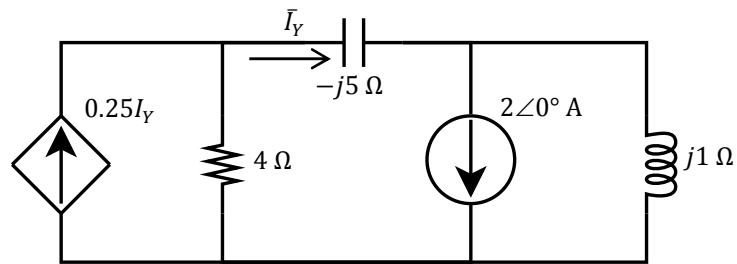


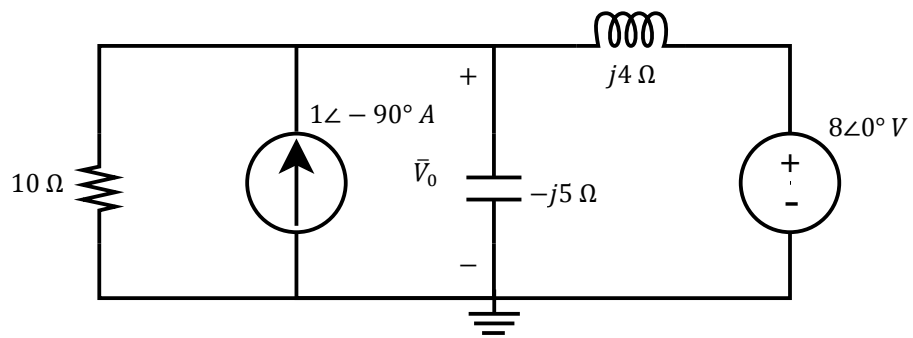
Vraag 4.5 [2]

Gebruik brontransformatie en bepaal $\bar{I}_Y$ in **reghoekige formaat**.

Question 4.5 [2]

Use source transformation and determine $\bar{I}_Y$ in **rectangular format**.



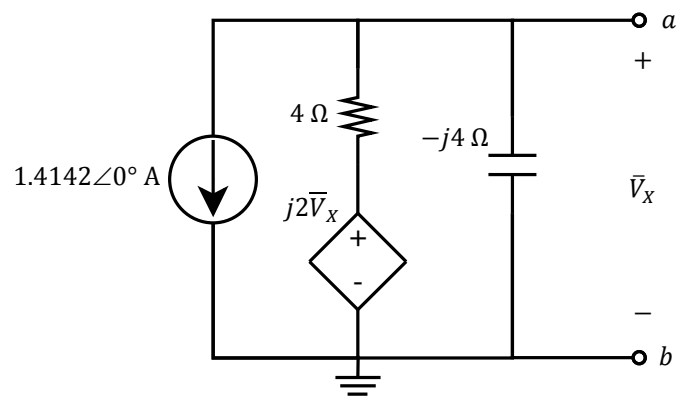
Vraag 4.6 [2]Bepaal $\bar{V}_0$ in reghoekige formaat.**Question 4.6 [2]**Determine $\bar{V}_0$ in rectangular format.

Vraag 4.7 [4]

Bepaal die Thevenin-ekwivalent tov terminale a - b . Skryf die spanningfasor in **polêre vorm** en die ekwivalente impedansie in **reghoekige vorm**.

Question 4.7 [4]

Find the Thevenin equivalent wrt terminals a - b . Write the voltage phasors in **polar form** and the equivalent impedance in **rectangular form**.



VRAAG 3 – QUESTION 3 (17)

3.1	$v_o =$		$i_o =$	
3.2	$a) R_f =$		$b) i_o =$	
3.3	$a) v_o =$		$b) R_I =$	
3.4	$v_I =$		$v_o =$	
3.5	$w_L =$		$w_C =$	
3.6	$a) v_C =$		$b) v_{L=5H} =$	

VRAAG 4 – QUESTION 4 (20)

4.1	a) $v_o(t) =$		
	b) _____ leads _____ by _____		
4.2	$\bar{I}_1 =$	(polar)	$\bar{V}_1 =$ (polar)
4.3	$i_x(t) =$		
4.4	$Z_{in} =$	(rectangular)	
4.5	$\bar{I}_Y =$	(rectangular)	
4.6	$\bar{V}_0 =$	(rectangular)	
4.7	$\bar{V}_{TH} =$	(polar)	$Z_{TH} =$ (rectangular)